Home Reading 10 — Optional PID Control

Before Class

Preview Lecture 10

Optional: read Lecture 10: PID Control.

Textbook Reading

- Ogata, System Dynamics, 4th edition, Section 10-3.

This Week Introduces

Proportional, integral and derivative actions, PI and PID design, saturation, windup and derivative filtering.

Tutorial Preparation

Read Tutorial 10 before class. Be ready to:

- Q1: Interpret controller actions and integral recovery after saturation.
- Q2: Assess a PI temperature controller with a finite heater.
- Q3: Audit two proposed PID settings.
- Q4: Explain how a derivative filter changes noise transmission and poles.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- Why can integral action remove steady-state error but cause windup during saturation?
- Why is matching a characteristic polynomial insufficient to guarantee acceptable performance?

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